

胶质瘤复发与坏死研究_阶段1

笔记本:	复发+坏死	更新时间:	2021/11/8 17:58
创建时间:	2021/11/8 14:14		
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数据集

- 原始数据分布
 - dicom格式的数据分布

```
"t1": {"R": 215, "N": 52, "RN": 82}
"t1ce": {"R": 223, "N": 53, "RN": 82}
"t2": {"R": 215, "N": 54, "RN": 93}
"flair": {"R": 215, "N": 24, "RN": 71}
```
- 数据清洗
 - 读取dcm文件信息，删除重复的影像信息，例如同个患者，即同个患者id，对应的study time和description等都一致
 - 疑问数据核对病理报告
 - 算法预处理
 - step1: 患者的相同StudyUID（同一次的检查数据）进行配准，4个模态配准，若该患者有T2则以T2为配准模板，若无，查找Flair, T1ce, T1
 - step2: 患者的相同StudyUID进行大小、像素空间等归一化，归一化模板与配准的模板一致
 - step3: 患者的颅骨提取，相同的StudyUID以配准模板的的颅骨为基准
 - step4: 人为检查，保证数据的准确性
 - 人为检查，删除不合适数据
 - 删除未进行化疗的数据：6个患者数据
 - 删除非胶质瘤患者数据：2个患者数据
 - 删除没有病理报告支持的数据：1个患者数据
 - 删除study的个别模态切片数较少：7个模态数据
 - 删除study的个别模态成像质量差的数据：2个模态数据
 - 删除study的颅骨去除效果差，破坏病灶组织的数据：7个患者数据（后续颅骨去除算法可继续优化，再保留使用）
- 清洗后数据分布
 - 310个患者数据，其中有22个患者有2次检测的数据,1个患者有3次检查数据，后续将多次检测的数据也视为独立的样本数据
 - 334个样本数据的类别分布情况如下，其中复发与坏死类别：[复发（R）](#)、[坏死（N）](#)、[复发&坏死（RN）](#)

(1) study患者的类别数量分布统计
注：当前数据中患者的多次检测的数据，对应的复发或坏死的类别都是一致

```
{
  'CHENG-SHUO_9000307252': [{'20200519': 'R'}, {'20190422': 'R'}],
  'ZHANG-JIAN-XIANG_9001394679': [{'20200622': 'R'}, {'20200621': 'R'}],
  'LI-FU-LIN_9000915543': [{'20190327': 'R'}, {'20190807': 'R'}],
  'SUN-DE-XING_9000449876': [{'20210118': 'R'}, {'20200203': 'R'}],
  'SHENG-PENG_0002721621': [{'20200110': 'R'}, {'20190806': 'R'}],
  'ZHOU-GANG_9000853419': [{'20200928': 'R'}, {'20200929': 'R'}],
  'LI-YANG_0002285915': [{'20190625': 'R'}, {'20190108': 'R'}],
  'LU-ZHEN-LI_0001794422': [{'20201124': 'R'}, {'20201202': 'R'}],
  'SONG-FENG-YING_0002186029': [{'20190402': 'R'}, {'20190723': 'R'}],
  'LV-AI-HONG_9001144568': [{'20190910': 'R'}, {'20191121': 'R'}],
  'YANG-GE-YUAN_9000571467': [{'20190125': 'R'}, {'20181026': 'R'}],
  'AN-JUN-HAI_9001270032': [{'20200709': 'R'}, {'20191226': 'R'}],
  'BIE-HAI_9001240646': [{'20200907': 'R'}, {'20210305': 'R'}, {'20191125': 'R'}],
  'MENG-FAN-ZHEN_9001463580': [{'20200814': 'R'}, {'20200813': 'R'}],
  'YAO-ZHI-HAO_637991': [{'20200123': 'R'}, {'20200122': 'R'}],
  'WEI-DONG_9000972610': [{'20200713': 'N'}, {'20200603': 'N'}],
  'HAO-AI-CUN_0002145406': [{'20200106': 'RN'}, {'20181210': 'RN'}],
  'ZHOU-LI-XIA_M694827': [{'20180327': 'RN'}, {'20171127': 'RN'}],
  'ZHANG-XU-NING_9001347530': [{'20201120': 'RN'}, {'20200420': 'RN'}],
  'WANG-GUI-HUA_9000610072': [{'20201222': 'RN'}, {'20181126': 'RN'}],
  'YAN-HONG-YUAN_9001231966': [{'20200922': 'RN'}, {'20201201': 'RN'}],
  'ZHAO-YA-LONG_9000149077': [{'20190514': 'RN'}, {'20210105': 'RN'}],
  'LEI-YIN-KU_M646117': [{'20180420': 'RN'}, {'20180522': 'RN'}]
}
```

study患者的类别数量分布统计

复发与坏死类别	R	N	RN	共计
样本数量	203	53	78	334

scan模态的数量分布统计

复发与坏死类别	T1	T1ce	T2	FLAIR	共计
R	188	190	191	174	743
N	51	50	53	23	177
RN	68	70	75	57	270

深度学习算法训练效果（类别分类）

注：用病灶分割算法获取的病灶区域标签，雨哲进行人为调整后的带有标注的标签信息还未清洗好，故还未采用这种方式训练效果

- 数据集划分1：T1+T1ce+T2+FLAIR，四种模态都包含（5折划分，训练集：验证集=4:1）

(1) 训练集

复发与坏死类别	T1	T1ce	T2	FLAIR	共计
R	152	154	155	142	603
N	39	38	41	20	138

(2) 验证集

复发与坏死类别	T1	T1ce	T2	FLAIR	共计
复发 (R)	38	38	38	34	148
坏死 (N)	12	12	12	3	39

- 数据集划分2：四种模态分别单独训练（5折划分，训练集：验证集=4:1）

影像模态	T1		T1ce		T2		FLAIR		共计
数据集	train	val	train	val	train	val	train	val	
复发 (R)	152	38	154	38	155	38	142	34	
坏死 (N)	39	12	38	12	41	12	20	3	
R+N总数	191	50	192	50	196	50	162	37	
R / (R+N)		76%		76%		76%		91.89%	

- 算法训练效果（注：训练一折的效果）

数据集类型	T1	T1ce	T2	FLAIR	T1+T1ce+T2	T1+T1ce+T2+FLAIR (bs=16)	T1+T1ce+T2+FLAIR (bs=8)
验证集 accuracy	86%	88%	88%	91.89% (86.49%)	83.33%	85.03%	89.30%
验证集 AUC	81.14%	85.09%	87.94%	33.33% (71.57%)	80.45%	-	90.12%

- 下图为对比论文中单模态数据样本的训练效果（Deep Learning Methodology for Differentiating Glioma Recurrence From Radiation Necrosis Using Multimodal Magnetic Resonance Imaging: Algorithm Development and Validation）

(1) 对比论文中样本数据分布

Table 1. Demographic and clinical data of the patient cohorts enrolled in this study.

Characteristic	Training set (n=117)	Test set (n=29)	Total (N=146)
Sample size (N=146), n (%)	117 (80.1)	29 (19.9)	146 (100)
Age in years, mean (SD)	40.9 (12.4)	42.0 (9.9)	41.1 (11.9)
Gender, n (%)			
Male	63 (53.8)	15 (52)	78 (53.4)
Female	54 (46.2)	14 (48)	68 (46.6)
Diagnosis of primary lesion, n (%)			
Grade II	33 (28.2)	8 (28)	41 (28.1)
Grade III	26 (22.2)	6 (21)	32 (21.9)
Grade IV	45 (38.5)	11 (38)	56 (38.4)
Unknown	13 (11.1)	4 (14)	17 (11.6)
Diagnosis of recurrent lesion, n (%)			
Necrosis	40 (34.2)	10 (34)	50 (34.2)
Glioma	77 (65.8)	19 (66)	96 (65.8)

(2) 对比论文中模型分类效果

Table 3. Performance of the deep neural network (DNN) models on individual magnetic resonance imaging (MRI) sequences: T1-weighted MRI (T1), T2-weighted MRI (T2), and gadolinium-contrast-enhanced T1-weighted MRI (T1c).

DNN model and magnetic resonance sequence	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy (95% CI)	Area under the curve (95% CI)
VGG^a16				
T1	0.725 (0.696-0.753)	0.606 (0.562-0.648)	0.684 (0.660-0.708)	0.718 (0.689-0.747)
T2	0.690 (0.660-0.719)	0.686 (0.644-0.727)	0.689 (0.665-0.713)	0.767 (0.740-0.794)
T1c	0.874 (0.851-0.894)	0.540 (0.496-0.585)	0.759 (0.736-0.781)	0.770 (0.743-0.797)
VGG19				
T1	0.804 (0.778-0.829)	0.448 (0.404-0.492)	0.681 (0.657-0.705)	0.692 (0.663-0.721)
T2	0.743 (0.714-0.770)	0.554 (0.510-0.598)	0.678 (0.653-0.702)	0.741 (0.713-0.769)
T1c	0.800 (0.773-0.825)	0.653 (0.610-0.694)	0.749 (0.726-0.771)	0.795 (0.769-0.821)
ResNet^b-50				
T1	0.782 (0.755-0.808)	0.584 (0.540-0.627)	0.714 (0.690-0.737)	0.732 (0.704-0.760)
T2	0.833 (0.808-0.852)	0.525 (0.480-0.569)	0.727 (0.703-0.750)	0.762 (0.735-0.789)
T1c	0.825 (0.799-0.848)	0.653 (0.610-0.694)	0.766 (0.743-0.787)	0.824 (0.800-0.848)
Inception-v3				
T1	0.724 (0.695-0.752)	0.596 (0.552-0.639)	0.680 (0.656-0.704)	0.706 (0.677-0.735)
T2	0.634 (0.603-0.665)	0.734 (0.693-0.772)	0.668 (0.644-0.693)	0.734 (0.706-0.762)
T1c	0.769 (0.741-0.795)	0.732 (0.691-0.770)	0.756 (0.733-0.778)	0.831 (0.807-0.855)
Inception-ResNet-v2				
T1	0.774 (0.746-0.800)	0.590 (0.546-0.633)	0.711 (0.687-0.734)	0.748 (0.720-0.776)
T2	0.829 (0.804-0.852)	0.529 (0.484-0.573)	0.726 (0.702-0.748)	0.804 (0.779-0.829)
T1c	0.812 (0.786-0.837)	0.722 (0.681-0.761)	0.781 (0.759-0.802)	0.841 (0.818-0.864)
ERN-Net^c				
T1	0.704 (0.674-0.732)	0.519 (0.474-0.563)	0.640 (0.615-0.665)	0.646 (0.615-0.676)
T2	0.634 (0.603-0.665)	0.606 (0.562-0.648)	0.624 (0.599-0.649)	0.675 (0.645-0.705)
T1c	0.803 (0.777-0.828)	0.643 (0.600-0.685)	0.748 (0.725-0.770)	0.807 (0.782-0.832)

^aVGG: Visual Geometry Group.^bResNet: residual neural network.^cERN-Net: efficient radionecrosis neural network.

